PEER REVIEW HISTORY

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ARTICLE DETAILS

TITLE (PROVISIONAL)	The effectiveness of educational interventions to develop patient safety knowledge, skills, behaviours, and attitudes in undergraduate nursing students: a systematic review protocol
AUTHORS	De Rezende, Helena; Vitorio, Aline; Morais, Alexandre; Garzin, Ana; Nicole, Andressa; Quadrado, Ellen; Lourenção, Daniela; Martins, Maristela

VERSION 1 – REVIEW

REVIEWER	Alhassan, Mustapha University for Development Studies, School of Allied Health Sciences
REVIEW RETURNED	02-Dec-2021

GENERAL COMMENTS	The dates of the study should be included in the manuscript, even though the Protocol was registered on the first of June 2021. The authors should report the planned date of the systematic review.

REVIEWER	Weller, Jennifer University of Auckland, Centre for Medical and health Sciences Education
REVIEW RETURNED	05-Dec-2021

GENERAL COMMENTS	Thank you for the opportunity to review this protocol. Patient safety is an important topic, and this review has the potential to promote better educational strategies in undergraduate nursing education. I do have a few comments and suggestions for the authors to consider. My main comments are around precision in the research question, the exclusion of interprofessional patient safety interventions, and no mention of how the contribution of qualitative researchers will be approached. Specific comments Patient safety is a very broad term – I'd suggest you could define it early on in your introduction, and justify why you chose patient safety topics as defined by WHO in their patient safety curriculum guide. It puts some boundaries around the topic early on – also helpful in the abstract. 6/40 mention made of nursing student learning needs as part of the reason to undertake this review – it's not clear how this review will provide information about nursing student learning needs – it seems the question is more about the effectiveness of educational interventions that teach a pre-defined set of topics. Consider deleting? The link https://www.who.int/teams/integrated-health services/patient-safety/policy/global-patient-safety-action-plan didn't work for me.

	The last few sentences of the introduction after the research question looks more like methods -consider moving the description of the search to methods? The term 'effectiveness' is rather vague – could you be more precise in the research question about the measures of effectiveness you will be looking for? Some universities offer post-graduate nursing programmes. Will these be included? How will qualitative research be treated – it has different measures for research quality and is unsuitable for a meta-analysis but may help develop a rich understanding of the effectiveness of an intervention, particularly a change in attitude, more difficult to assess with measurement tools. 9 / 12 under Outcomes – how will you make decisions or not on whether and evaluation instrument is 'validated'? Validity is not a black and white concept but a continuum – you build evidence on validity, and that validity is specific to the context in which the instrument has been tested. I wondered about the decision to exclude interprofessional interventions - teamwork is one of the WHO topics – healthcare is delivered by teams of nurses, doctors and allied health professionals. I'm aware of undergraduate patient safety interventions that are purposefully interprofessional as a key component of their educational approach. Your review might exclude some of the more useful patient safety educational interventions. Any opportunity to rethink this? Why exclude studies that are not in English, Spanish or Portuguese? Of included studies two methods are described – a meta-analysis of studies where possible and a narrative synthesis of other studies. How will these two approaches be integrated to provide an evidence-based study outcome? Will the meta-analysis be privileged?

VERSION 1 – AUTHOR RESPONSE

Reviewer 1: Dr. Mustapha Alhassan, University for Development Studies
The dates of the study should be included in the manuscript, even though the Protocol was registered on the first of June 2021.
RESPONSE: Dear reviewer, we greatly appreciate the thoughtful comments that helped improve the manuscript and thank you for pointing that out. We have added the study dates in the Abstract (please see marked copy page 2, lines 45-46) and Methods (page 10, line 212).

The authors should report the planned date of the systematic review
RESPONSE: Thank you for indicating this. The planned date of the systematic review is June 2022. This information was added to the methods (Page 6 line 153).

Reviewer 2: Dr. Jennifer Weller, University of Auckland
I do have a few comments and suggestions for the authors to consider.
• My main comments are around precision in the research question, the exclusion of interprofessional patient safety interventions, and no mention of how the contribution of qualitative researchers will be approached.

Patient safety is a very broad term – I'd suggest you could define it early on in your introduction, and justify why you chose patient safety topics as defined by WHO in their patient safety curriculum guide. It puts some boundaries around the topic early on – also helpful in the abstract.
RESPONSE: Dear reviewer, thank you for your careful review of the manuscript; your comments and questions have helped us to improve the current version of our paper. A definition of patient safety was added at the beginning of the introduction (please see marked copy page 4, lines 74-77). The rationale for choosing the WHO patient safety curriculum was provided in the search strategy section; hence, we moved it to the introduction (please see page 6, line 140-145) . These aspects were also included in the abstract as per your suggestion (page 2, lines 32-33 and lines 39-40).

6/40 mention made of nursing student learning needs as part of the reason to undertake this review – it's not clear how this review will provide information about nursing student learning needs – it seems the question is more about the effectiveness of educational interventions that teach a pre-defined set of topics. Consider deleting?
RESPONSE: Thank you for your suggestion. We have considered it pertinent, and we have deleted this sentence.

The link https://www.who.int/teams/integrated-health services/patient-safety/policy/global-patient-safety-action-plan didn't work for me. Review. https://www.who.int/teams/integrated-health-services/patient-safety/policy/global-patient-safety-action-plan
RESPONSE: Thank you for indicating this. We have reviewed the link, and despite being correct, we have added it again. It should work now (please see page 15, line 335, reference number 1).

The last few sentences of the introduction after the research question looks more like methods - consider moving the description of the search to methods?
RESPONSE: Thank you for the suggestion. We have removed those sentences. We did not move them to the methods because that information was already provided in the methods section.

The term 'effectiveness' is rather vague – could you be more precise in the research question about the measures of effectiveness you will be looking for?
RESPONSE: Thank you for pointing that out. We have reviewed the research question and have replaced the term effectiveness to make it more precise in order to identify what we want to measure: the impact of educational interventions (please see page 7, lines 158-159).

Some universities offer post-graduate nursing programmes. Will these be included?
RESPONSE: These programmes will not be included. It is suggested that patient safety training needs to start early in the undergraduate nursing programmes. Hence, we want to investigate the effectiveness of educational interventions to teach patient safety for this population.

How will qualitative research be treated – it has different measures for research quality and is unsuitable for a meta-analysis but may help develop a rich understanding of the effectiveness of an intervention, particularly a change in attitude, more difficult to assess with measurement tools.
RESPONSE: As much as we found your suggestion very pertinent, we do not intend to include and/or analyse qualitative studies as a systematic review of effectiveness. This would fit better into a new project of our group since it could change the direction of our focus on this paper.

9 / 12 under Outcomes – how will you make decisions or not on whether and evaluation instrument is 'validated'? Validity is not a black and white concept but a continuum – you build evidence on validity, and that validity is specific to the context in which the instrument has been tested.
RESPONSE: Thank you for your insightful comment. We agree; therefore, we have removed this information from the outcomes section. We will consider all studies that described and evaluated at least one of the following outcomes: nursing students' knowledge, skills, attitudes, and behaviours

related to patient safety. However, we will also extract data regarding the reliability and validity of the instruments used to evaluate the intervention as outlined by Reed et al., 2005 (Please see marked copy page 8, lines 186-188); additionally an information was added to the data extraction section (pages 11-12, lines 263-265) and Appendix 3 –Extraction Tool.

I wondered about the decision to exclude interprofessional interventions - teamwork is one of the WHO topics – healthcare is delivered by teams of nurses, doctors and allied health professionals. I'm aware of undergraduate patient safety interventions that are purposefully interprofessional as a key component of their educational approach. Your review might exclude some of the more useful patient safety educational interventions. Any opportunity to rethink this?
RESPONSE: We agree that this may exclude important patient safety interventions. However, we decided to investigate only undergraduate nursing students as we want to expand the knowledge of the teaching of effective patient safety educational interventions to this group. A future project about interprofessional patient safety education would be more appropriate since it could alter the direction of this research.

Why exclude studies that are not in English, Spanish or Portuguese?
RESPONSE: Thank you for highlighting this. We have included studies in English, Spanish and Portuguese because these are the languages of the systematic review team. We have added this information to that section of the manuscript. We are aware that this decision can pose risk of bias to the manuscript; however, the review team does not have capacity to translate studies published in languages other than those stated. In order to address this, we will include studies published in other languages if an English translation is available (please see 10, lines 209-211).

Of included studies two methods are described – a meta-analysis of studies where possible and a narrative synthesis of other studies. How will these two approaches be integrated to provide an evidence-based study outcome? Will the meta-analysis be privileged?
RESPONSE: Thank you for emphasising this part. We will be privileged meta-analysis. However, we are aware that this may not be feasible due to heterogeneity of the studies of educational interventions, as stated in the synthesis section. If meta-analysis is not possible, the data will be presented narratively. They will follow the Synthesis without meta-analysis (SWiM) in systematic reviews: reporting guideline (Campbell et al., 2020) as cited in the last paragraph of the synthesis section to integrate these two approaches (please see page 12, lines 289-291)

VERSION 2 – REVIEW

REVIEWER	Weller, Jennifer University of Auckland, Centre for Medical and health Sciences Education
REVIEW RETURNED	05-Feb-2022

GENERAL COMMENTS	The authors have addressed my comments. I don't have any further suggestions.